

HubForge OS Knowledge Graph

Purpose

HubForge OS depends on structured knowledge systems.

The system cannot reason effectively without organized expert knowledge.

Knowledge must be machine-readable.

This document defines the knowledge ontology.

Core Knowledge Architecture

HubForge OS organizes knowledge into eight major layers.

Layer 1 — Domain Knowledge

Layer 2 — Framework Knowledge

Layer 3 — Procedural Knowledge

Layer 4 — Decision Rules

Layer 5 — Evidence Libraries

Layer 6 — Historical Memory

Layer 7 — Reasoning Patterns

Layer 8 — Improvement Heuristics

Domain Knowledge Layer

Contains specialized domain expertise.

Examples:

Social Impact

Healthcare

Education

Climate Science

Law

Public Policy

Research Systems

Enterprise Strategy

Framework Knowledge Layer

Structured frameworks experts use repeatedly.

Examples:

Social Impact

- Theory of Change
- Logframes
- Outcome Mapping
- Evaluation Frameworks

Healthcare

- Diagnostic Protocols
- Clinical Guidelines
- Treatment Trees

Research

- Experimental Design
- Statistical Frameworks
- Survey Methodology

Policy

- Policy Simulation Frameworks
- Governance Evaluation Models

Legal

- Case Reasoning Structures
- Compliance Frameworks

Procedural Knowledge Layer

Stores process knowledge.

Example:

Research Process

Question Formulation

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Literature Review

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Hypothesis Design

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Method Selection

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Sampling Design

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Data Collection

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Analysis

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Interpretation

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Publication

This procedural logic becomes machine executable.

Decision Rules Layer

Deterministic rules.

No AI required.

Examples:

SMART Goal Validation

If goal has no measurable target

Reject

Survey Validation

If sample size too small

Flag error

Healthcare Rules

If symptom cluster incomplete

Require additional data

Policy Rules

If evidence confidence low

Require verification

Evidence Library Layer

Contains evidence sources.

Examples:

Research Papers

Policy Reports

Institutional Frameworks

Government Guidelines

Historical Cases

Structured Databases

Domain Documentation

Historical Memory Layer

Stores prior reasoning outcomes.

Contains:

- Previous decisions
- Successful solutions
- Failed solutions
- Institutional knowledge
- Repeated reasoning patterns

This prevents repeated mistakes.

Reasoning Patterns Layer

Stores expert reasoning templates.

Examples:

Diagnosis Reasoning

Hypothesis Testing

Comparative Analysis

Failure Analysis

Root Cause Analysis

Counterfactual Reasoning

Risk Modeling

Improvement Heuristics Layer

Stores methods for improving reasoning quality.

Examples:

Find weak assumptions

Detect logical inconsistency

Detect missing evidence

Replace vague outputs

Strengthen measurable targets

Reduce uncertainty

Improve causal logic

Core Principle

Knowledge must become infrastructure.

Expertise should be structured, reusable, and machine readable.

This becomes the intelligence foundation of HubForge OS.